👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 4 Compression (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 3: Compression Handout + Compression Article”)	Level 1 piece
Compression Handout (.doc)	Deck + notes A-C + quiz
Compression Article (scanned image)	Level 3 reading replaces it (our own text)
Class slides	Deck: 13 slides, Words to Know first, “point to the picture” checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Demo: squeeze a syringe or bike pump with a finger over the end to feel compression

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part
- Cover one choice to make it 2

4. What to grade

- Compression Handout: L1 notes + quiz; L2 notes + quiz; same points
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Compression Article**: The article is a scanned image (source unknown); our reading covers the same ideas.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Compression Test (Level 1)

Small Engines Overhaul · Unit 4 · goes with the class “Compression Handout” (lab points). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Compression Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Model first, then the student tries with adult supervision
- Picture task card at the bench
- Parts tray with labeled cups for small parts

4. What to grade (lab points)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Compression spec depends on the engines in Mr. Davis's shop; staff fill in the spec.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

💨 Compression Notes

Small Engines Overhaul · Unit 4 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

 squeeze _____

 rings _____

 leak _____

 gauge _____

🖼️ B. Fill in the word. Use the pictures.

5.  Compression means _____.

6.  Piston _____ seal.

7.  A _____ makes it weak.

8.  The _____ reads the number.

Word bank:  **gauge**  **leak**  **rings**  **squeeze**

⭕ C. Circle the right picture.

9. Good squeeze makes…

 Sleep	 Power

10. The gauge goes in the…

 Gas tank	 Spark plug hole

11. Worn rings…

 Leak	 Shine

🔗 D. Draw a line from the picture to the word.

	valve
	gasket
	gauge
	rings

📖 Compression: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Compression

squeezing the air and gas



Piston rings

rings that seal the piston



Valve seat

where the valve closes



Head gasket

seal between head and block



Compression gauge

tool that reads squeeze



PSI

pounds per square inch



Leak

air gets out



Low compression

weak squeeze

Compression Test

Small Engines Overhaul · Unit 4 · Level 1

Name: _____

Modified from the class “Compression Handout”.

Supplies: get it. Check ☐.

- ☐ **Compression gauge** with adapter
- ☐ **Spark plug socket**
- ☐ **Practice engine**
- ☐ **Oil can** for wet test, L2/L3
- ☐ **Data sheet**

📟 Do one box. Check ☐ it.

1		Glasses on.	☐
2		Plug wire off.	☐
3		Take out the spark plug with an adult.	☐
4		Screw in the gauge.	☐
5		Pull the rope 5 times.	☐
6		Read the number with an adult.	☐
7		Put the plug back.	☐

🚧 **Under construction:** Compression spec depends on the engines in Mr. Davis's shop; staff fill in the spec. *Check with the class teacher.*

📟 Compression Test · Get Ready

Small Engines Overhaul · Unit 4 · Level 1

Name: _____

👥 My job. Circle it.

- **Mechanic** — fixes
- **Tool helper** — gets tools
- **Recorder** — writes
- **Clean-up captain** — cleans

🧰 Get the tools. Check ☐.

- ☐ Compression gauge
- ☐ Spark plug socket
- ☐ Practice engine
- ☐ Oil can
- ☐ Data sheet

⚠️ Be safe!

- Glasses on.
- Coveralls on.
- Tools with an adult.
- Face away.
- Clean up, wash hands.

✅ Before we start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

📟 Compression Test · My Check

Small Engines Overhaul · Unit 4 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🥽 I wore my safety gear
- ☐ 🔧 I did the engine job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🥽 **I was safe.**	😀	🙂	😕
🔧 **I did my job.**	😀	🙂	😕
⭐ **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Compression Quiz

Small Engines Overhaul · Unit 4 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🗜️ Compression is…

- 🧊 Freezing
- 🗜️ Squeezing

2. ⭕ Rings…

- ⭕ Seal the piston
- 🎵 Make music

3. 🚪 Valves must…

- 🚪 Close tight
-  Stay open

4. 💨 A leak makes it…

- 💪 Strong
-  Weak

5. 📟 Reads pressure…

-  Ruler
- 📟 Gauge

6. Gauge goes in the…

- Plug hole
- Tank

7. Seals head to block…

- Paper towel
- Head gasket

8. Pressure unit…

- PSI
- Degrees

💨 Compression Notes · KEY

Small Engines Overhaul · Unit 4 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

 squeeze _____

 rings _____

 leak _____

 gauge _____

🖼️ B. Fill in the word. Use the pictures.

5.  Compression means **squeeze**.

6.  Piston **rings** seal.

7.  A **leak** makes it weak.

8.  The **gauge** reads the number.

Word bank:  **gauge** |  **leak** |  **rings** |  **squeeze**

⭕ C. Circle the right picture.

9. Good squeeze makes…

- 😴 Sleep
- **💪 Power** ✓

10. The gauge goes in the…

- ⛽ Gas tank
- **🔩 Spark plug hole** ✓

11. Worn rings…

- **💨 Leak** ✓
- 🌟 Shine

🔗 D. Draw a line from the picture to the word.

Key: ⭕ → rings · 🚪 → valve · 🧾 → gasket · 📟 → gauge

⭕	valve
🚪	gasket
🧾	gauge
📟	rings

🔑 *Key written by our team from the class teacher's materials, please check.*

✅ Compression Quiz · KEY

Small Engines Overhaul · Unit 4 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🗜️ Compression is… Compression Handout

- Freezing 
- **Squeezing** ✓ 

2. ⭕ Rings… Compression Handout

- **Seal the piston** ✓ 
- Make music 

3. 🚪 Valves must… Compression Handout

- **Close tight** ✓ 
- Stay open 

4. 💨 A leak makes it… Compression Handout

- Strong 
- **Weak** ✓ 

5. 📟 Reads pressure… Compression Handout

- Ruler 
- **Gauge** ✓ 

6. Gauge goes in the… Notes

- **Plug hole**
- **Tank**

7. Seals head to block… Notes

- **Paper towel**
- **Head gasket**

8. Pressure unit… Notes

- **PSI**
- **Degrees**

Answers

1-B (Squeezing) · 2-A (Seal the piston) · 3-A (Close tight) · 4-B (Weak) · 5-B (Gauge) · 6-A (Plug hole) · 7-B (Head gasket) · 8-A (PSI)

Level 1 version of the gen ed “Compression Handout”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*